Cell Culture (Lecture)

Course ID: BIOT 780/880
Term Start and End Dates: Fall 2023 - August 28th, 2023 – December 11th, 2023.
Location and Times: Lecture – P345, MW 8:40am-10:00;
Morning Lab – P650EC, MW 10:10am-12:30pm;
Evening Lab – P650EC, MW 6:40pm - 9:00pm
Instructor name: Mark T. Scimone, PhD (Lecture and morning lab section)
Evening Lab Instructor: Matthew Woodworth
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Office: Room 628
Office Phone: 603-641-4389
Office Hours: Mondays 2-3pm

COURSE DESCRIPTION

Principles and technical skills fundamental to the culture of animal and plant cells, tissues, and organs. Introduction to the techniques of sub-culturing, establishing primary cultures, karyotyping, serum testing, cloning, growth curves, cryopreservation, hybridoma formation and monoclonal antibody production, and organ cultures. Application of cell culture to contemporary research in the biological sciences.

COURSE OBJECTIVES. *We aim to teach:*

1. The history, theory, and biology of mammalian cell culture.
2. Biotechnology and basic research applications of eukaryotic (and prokaryotic) cell culture.
3. Aseptic technique and cell culture assays important to basic research and biotechnology.
4. Science communication through oral presentations, grant/project proposal writing, and lab reports.
5. Critical thinking and problem solving through evaluation of real data sets incorporating cell culture techniques.

STUDENT LEARNING OUTCOMES

After completing the cell culture course, students should be able to:

1. Understand the history, theory and biology behind the cell culture techniques and reagents utilized in the laboratory.
2. Aseptically culture mammalian cells.
3. Learn to use equipment properly to carry out a variety of cell culture techniques, including cell subculturing, viability, cytotoxicity, cryopreservation, transfection, immunofluorescence, flow cytometry, etc.
4. Critically evaluate data from primary research articles to understand the application of classroom learning to current basic research fields.

5. Act like researchers by developing a project plan, writing a grant proposal, carrying out experiments, statistically analyzing data, using graphing software to display data appropriately, and present findings to an audience.

Textbook

Online Textbook. Please check Canvas for details.

Course Structure

Canvas is the learning management tool we use for this course. (AKA MyCourses) modules will contain: Power Point presentations, all readings, useful websites, and online assignments.

Course Navigation	*Description*
Home Page	Landing Page for the course. Links to individual modules.
Syllabus	The syllabus, course schedule and other key class documents are located here.
Modules	This area contains the learning Modules (each week is its own module). The course content, activities, and assignments are located in this area.

Course Schedule

Week	Date	Lecture
1	Aug 28 Aug 30	Introduction to Course; Terminology, Basic cell culture
2	Sep 6	Sep 4th is Labor Day (University Holiday) Biology of Cultured Cells & Cell Characterization
3	Sep 11 Sep 13	Environmental Factors/Nutrition
4	Sep 18 Sep 22	Quantifying Cell Growth and Other Procedures **Quiz #1 (Sep 22)**
5	Sep 25 Sep 27	Studying Cancer Cells: Identifying cell growth and attachment problems, etc.
6	Oct 2 Oct 4	
7	Oct 9 Oct 11	October 9th is Fall Break (No classes) **MIDTERM EXAM (Oct 11)**
8	Oct 16 Oct 18	Gene Delivery & Cloning
9	Oct 23 Oct 25	Cell Phenotyping
10	Oct 30 Nov 1	3-D cell Culture **Quiz #2 (Nov 1)**
11	Nov 6 Nov 8	Mammalian Cell Culture Industry/Manufacturing Applications
12	Nov 13 Nov 15	Site visit or guest lectures from industry (TBD)

13	Nov 20 Nov 22	Cutting Edge Approaches in Cell Culture
14	Nov 27 Nov 29	Flex day (Snow Day accommodations)
15	Dec 4 Dec 6	**FINAL EXAM (Dec 6)**

Methods of Testing/Evaluation: The lecture course grade will be based on following:

Method of Evaluation	Total Point Value
2 Exams* - 150 pts ea	300
Weekly in-class or homework assessments – 15 pts ea	150
2 Quizzes – 75 pts ea	150
TOTAL POINTS AVAILABLE	600

*Note: Additional questions will be added to each exam for 800/graduate-level students. For graduate students, your overall BIOT 853 grade will be determined by averaging the lecture and lab grades (50%/50%).

Course Assessments

- Exams: A midterm exam (October 11th) and a final exam (December 6th) will be given for this course. Both exams will cover material from lecture and the background and protocols from experiments carried out in lab. The format of these exams will largely be short answer/essay, and questions will be like those in the weekly in-class/homework assignments.
- Quizzes: Two 45-minute quizzes will be given during the semester - during Wednesday lecture periods (Sep 22, Nov 1). Both quizzes will cover material from lecture and the experiments carried out in lab. These quizzes will contain a variety of formats: fill-in-the-blank, matching, true-false, multiple choice, and short answer.
- Weekly Homework Questions / In-class Assignments: To support the application of the content learned through lectures, you will be assigned either short answer questions online (posted Monday pm or Tuesday am) pertaining to the week's lecture and reading (and sometimes lab material) or work on individual or collaborative assignments during Wednesday lecture classes. For online assignments, questions are due each week by Sunday, 11:59 pm. Late assignments will not be accepted.

Class Attendance

Attendance is required.

Lab/Classroom Conduct

- *Laboratory Conduct:* You must wear long pants or a long skirt, wear close-toed shoes, and pull any hair back in the lab.
- *Cell phones:* Please **turn off/silence** and put away your cell phone before you enter the classroom.

- *Computer Use:* You may use a laptop *during lab/lecture* for taking notes ONLY. You should NOT be using your computer for social networking, emailing, playing games, etc. Such behavior distracts you and your classmates from the lecture and is considered disrespectful to the professor and your classmates. If you are judged to be misusing your computer in any way, you will be asked to put it away for the remainder of the lecture.
- *Food:* There is a strict policy of NO food or drinks in the laboratory. However, you may choose to have drinks in the classroom.

Make-up Quizzes/Exams

Exams and quizzes are given only on the announced date. Make-up exams and quizzes will be given ONLY if it is missed for a valid (documented) reason and discussed ahead of time.

Communication Expectations

I will be active in the canvas page daily, Monday through Friday. If you post a question for me in a forum or if you email me, anticipate a response within 24-hours or sooner during weekdays. On the weekends I may not log in at a regular time. If you write to me over the weekend, I may not respond until Monday morning.

How to Reach Me

Questions related to assignments or learning should first be asked, if possible, through canvas. Otherwise, feel free to email me or come to office hours.

Academic Integrity

Unless otherwise specified, the use of Automated Writing Tools, including artificial intelligence (AI) tools, is strictly prohibited in this course, even when properly attributed. The use of automated writing tools is considered plagiarism (as defined by UNH's Academic Integrity Policy) and will be handled in accordance with existing policy. Students are expected to review these policies.

https://catalog.unh.edu/srrr/university-policies-regulations/academic-honesty/

PLAGIARISM (from SRRR)

Use or submission of intellectual property, ideas, evidence produced by another person, including computer generated text or work outsourced to third-parties, in whole or in part as one's own in any academic assessment without providing proper citation or attribution. In some cases, reusing one's own previous work without acknowledging or citing the original work can constitute self-plagiarism.

Note: This syllabus is subject to change. Students will be promptly notified of any changes.

UNH Manchester Campus Support

Library: The UNH Manchester librarians are available to assist you with your research. You can contact a librarian by calling 603-641-4173 or by emailing unhm.library@unh.edu.
The following online resources provide information about library resources and services:
• UNH Manchester Library webpage: https://manchester.unh.edu/library
• Online Research Guides: https://libraryguides.unh.edu/index.php?b=s
• Access Library Resources Remotely: https://libraryguides.unh.edu/remoteaccess
• Reserve a study room for Zoom classes:
https://libraryguides.unh.edu/remoteaccess/studyrooms

The Center for Academic Enrichment (CAE) professionals and peers are available to support all UNH Manchester students in maximizing their learning potential through individual in-person and online tutoring, in-class workshops, and study groups in math, writing, course content, study skills, time management, and personal statements. All students registered for UNH Manchester courses are entitled to one hour of individual tutoring, per course, per week. Appointments are available at https://caetutor.unh.edu; for more information, contact the CAE at (603) 641-4113, or unhm.cae@unh.edu .

Student Accessibility Services (SAS)
According to the Americans with Disabilities Act (as amended, 2008), each student with a disability has the right to request services from UNH to accommodate his/her/their disability. If you are a student with a documented disability or believe you may have a disability that requires accommodations, please contact Student Accessibility Services (SAS) located on the Manchester campus in the Student Services Suite (Office 405A). Accommodation letters are created by SAS with the student. Please follow-up with your instructor as soon as possible to ensure timely implementation of the identified accommodations in the letter. Faculty have an obligation to respond once they receive official notice of accommodations from SAS but are under no obligation to provide retroactive accommodations.
For more information refer to www.unh.edu/sas or contact SAS at 603.862.2607, 711 (Relay NH) or sas.office@unh.edu.

Emotional or mental health distress
In partnership with The Mental Health Center of Greater Manchester, UNH Manchester offers consultation visits on a walk-in basis and through telehealth appointments. Services include:

- Free confidential screening & consultation with a licensed mental health therapist.
- Referrals to mental health or substance misuse treatment. And assistance in understanding how to afford additional treatment (with or without insurance!) or find free services.

You may email: unhm.wellness@unh.edu to make an appointment to meet with a counselor by clicking here or by using the QR codes below.

For in person appointments please scan this code

For remote appointments please scan this code

If you would like to connect to counseling services directly, you may do so by contacting The Greater Manchester Mental Health Center at (603) 668 - 4111.
The National Suicide Prevention Lifeline provides 24/7, free and confidential support via phone or chat for people in distress, resources for you or your loved ones, and best practices for professionals. Call (800) 273-TALK (8255).

Sexual Harassment and Rape Prevention Program (SHARPP): Provides free and confidential advocacy and direct services to survivors. (https://www.unh.edu/sharpp.)
UNH Manchester's SHARPP Office Hours during Fall & Spring Semesters are Mondays 9am-4pm in Room 471. Zoom Appointment Availability year round is Mon-Fri 9am-4pm
24/7 Crisis Line: 603-862-SAFE (7233)
Main Office: 603-862-3494
UNH Manchester students can also contact the YWCA of New Hampshire – 603-668-2299 (24hour), 72 Concord St. Manchester, NH, for crisis or emergency services.

Basic Needs Support: Food, Housing, Financial. https://www.unh.edu/dean-of-students/getting-help/housing-food-financial-basic-needs-support

Food Pantry The UNH Manchester Food Pantry, located in room 437 is open Monday through Friday from 8:00am- 9:30pm. Any UNH Manchester community member can take what they need. If you have any questions please email UNHM.Foodpantry@unh.edu